

VEVAN O NARAIN

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Third-year B.Tech CSE (AI/ML) student at VIT Vellore (9.44 GPA) building production-grade systems spanning real-time collaborative infrastructure and applied ML/retrieval — from CRDT-based collaborative editors to hybrid-search RAG pipelines. Seeking an SDE internship.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, C++, SQL
- **Web Development:** React.js, Node.js, Express.js, PostgreSQL, MongoDB, REST APIs, Redis
- **AI/ML:** Scikit-learn, TensorFlow, PyTorch, Sentence Transformers, ChromaDB, RAG
- **Developer Tools:** Git, GitHub, Docker, VS Code, Jupyter Notebook
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, Vector Search

PROJECTS

Vync - Real-Time Collaborative Editor [*React, Node.js, Socket.IO, Yjs, PostgreSQL, Redis, Docker*] [GitHub](#)

- Built a Google Docs-style collaborative code editor with CRDT-based (Yjs) real-time synchronization across six programming languages using WebSockets for low-latency multi-client updates.
- Engineered conflict-free state reconciliation ensuring consistency across concurrent edits without merge conflicts.
- Sandboxed multi-language code execution in Docker containers with CPU and memory limits for secure, isolated execution.

10Kwery - SEC Filing RAG System [*Python, ChromaDB, Sentence Transformers, BM25, Groq API, BeautifulSoup*] [GitHub](#)

- Built an end-to-end Retrieval-Augmented Generation (RAG) system for querying SEC 10-K filings using natural language, returning citation-grounded answers linked to source excerpts.
- Designed a hybrid retrieval pipeline combining dense vector search, BM25 keyword retrieval, Reciprocal Rank Fusion (RRF), and cross-encoder reranking to improve retrieval precision.
- Built an evaluation framework measuring MRR, Hit@1, Hit@5, and citation validity to benchmark retrieval quality and drive pipeline improvements.

Plant Leaf Disease Detection [*Python, TensorFlow, Keras, ResNet50, Streamlit, Scikit-learn*] [GitHub](#)

- Fine-tuned ResNet50 using transfer learning on a dataset of over 50,000 plant leaf images, achieving 98.75% accuracy across 38 disease classes.
- Deployed a Streamlit application for real-time image upload and disease classification with prediction confidence scores.

Real-Time Emotion Detector [*Python, TensorFlow/Keras, OpenCV, CNN, NumPy*] [GitHub](#)

- Built a real-time facial emotion recognition system using OpenCV and a CNN trained on the FER-2013 dataset containing over 30,000 facial images.
- Classified seven facial emotions from live webcam input and applied image preprocessing and data augmentation to improve robustness across varying lighting and pose conditions.

EDUCATION

• **B.Tech in Computer Science and Engineering (AI & ML Specialization)**

Vellore Institute of Technology, Vellore

Cumulative GPA: 9.44 | Expected Graduation: May, 2028

• **Senior Secondary (Class XII)** — Modern School Barakhamba Road, CBSE Board

Percentage: 90.2% (PCM with CS) | Completed: May, 2024

• **Secondary (Class X)** — Modern School Barakhamba Road, CBSE Board

Percentage: 96% | Completed: May, 2022

CERTIFICATIONS & ACHIEVEMENTS

- **Machine Learning A-Z** by SuperDataScience Team
- **The MERN Fullstack Guide** by Maximilian Schwarzmüller and Manuel Lorenz
- Solved 200+ problems on LeetCode, strengthening problem-solving skills and proficiency in data structures and algorithms.